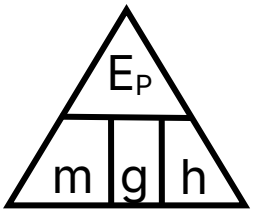
Calculating Potential Energy, E_P

The potential energy of an object is dependent on 3 factors: the mass (m) of the object, gravity and its height. The formula for calculating kinetic energy is:



E_P= mgh

Potential energy = mass x gravity x height

- Mass of the object is measured in kg
- Gravity is measured in m/s^2 or ms^{-2} (either one is accepted as they are the same unit of measurement)
- Height is measured in m

Example

A drone has mass of 3kg and flies to an altitude of 125m. Calculate its potential energy.

E_P = mgh

= 3 x 9.8 x 125

= 3,675 J

= <u>3.675 kJ</u>

What are my knowns?
m = 3kg h = 125m
g = 9.8ms^{-2}



Questions

(Calculate answer to two significant figures)

1. A 1.5kg toy buggy is set up to roll down ramps of different height to test its potential energy. Calculate how much energy the buggy possesses at the top of each of the following ramp heights:

   a) 50 cm

   E_P = mgh

   E_P = 1.5 x 9.8 x 0.5

   E_P = 7.35 J = **7.4 J** to two significant figures

   b) 75 cm

   E_P = mgh

   E_P = 1.5 x 9.8 x 0.75

   E_P = 11.025 J = **11 J** to two significant figures

   c) 1 m

   E_P = mgh

   E_P = 1.5 x 9.8 x 1

   E_P = 14.7 J = **15 J** to two significant figures

   d) 1.24 m

   E_P = mgh

   E_P = 1.5 x 9.8 x 1.24

   E_P = 18.228 J = **18 J** to two significant figures

2. Using a ramp height of 1.22m, calculate the potential energy of the buggy with different weights.

   a) 2.5 kg

   E_P = mgh

   E_P = 2.5 x 9.8 x 1.22

E_P = 29.89 J = **30 J** to two significant figures

b) 3.5 kg

E_P = mgh

E_P = 3.5 x 9.8 x 1.22

E_P = 41.846 J = **42 J** to two significant figures

c) 6kg

E_P = mgh

E_P = 6 x 9.8 x 1.22

E_P = 71.736 J = **72 J** to two significant figures

d) 400g

E_P = mgh

E_P = 0.4 x 9.8 x 1.22

E_P = 4.7824 J = **4.8 J** to two significant figures

3. A roller coaster has many high and low points on its track. The height at the top of its largest dip is 50m, and the lowest point is 5m, what is the change in potential energy for a person with a mass 80kg between the highest and lowest points?

E_{P1} = mgh

E_{P1} = 80 x 9.8 x 50

E_{P1} = 39,200 J

E_{P2} = mgh

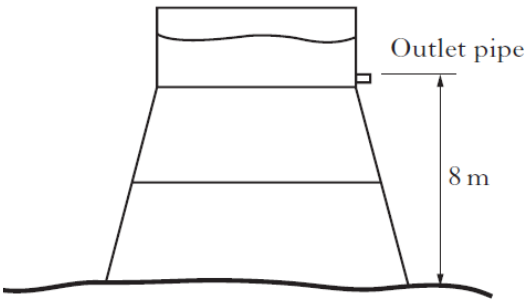
E_{P2} = 80 x 9.8 x 5

E_{P2} = 3,920 J

Therefore, change in potential energy, ΔE_P, is:

ΔE_P = 39,200 – 3,920 = 35,280 J = **35 kJ** to two significant figures

4. A crofter uses a wind power system to pump water to a storage tank. A simplified diagram of the storage tank is shown.



a) Calculate the potential energy of 6L (litres) of water at the outlet pipe (1L of water = 1 kg).

E_P = mgh

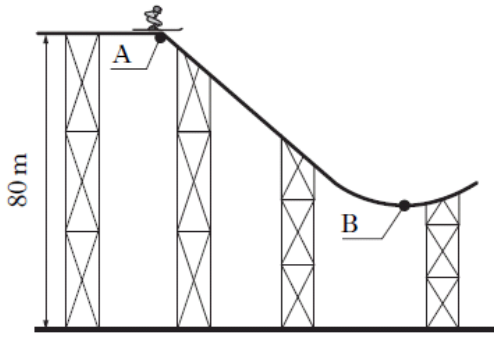
E_P = 6 x 9.8 x 8

E_P = 470.4 J = **470 J** to two significant figures

b) Describe why not all of the water's potential energy is converted.

Energy lost due to sound, heat (caused by friction of liquid against pipe internals).

5. A ski jumper and take-off ramp is shown. The ski jumper has a mass of 65kg.



a) Calculate, showing all working and units, the potential energy of the ski jumper when stationary at point A.

E_P = mgh

E_P = 65 x 9.8 x 80

E_P = 50,960 J = **51kJ** to two significant figures

b) The ski jumper descends down the ramp and has a potential energy of 31.5 kJ at point B.

(i) Calculate the kinetic energy of the ski jumper at point B (assume no loss of energy).

Therefore, potential energy converts only to kinetic energy.

So the lost potential energy from point A to B must all be converted to kinetic energy at point B.

(Note: the skier has a mixture of potential and kinetic energy at point B – they are moving and also at a height)

From 5a), E_{PA} = 50,960 J At point B, E_{PB} = 31.5kJ = 31,500 J

$E_{KB} = E_{PA} - E_{PB}$

E_{KB} = 50,960 – 31,500

E_{KB} = 19,460 J = **19 kJ** to two significant figures

(ii) The velocity of the ski jumper at point B.

$E_{KB} = \frac{1}{2}mv^2$

$v^2 = (2 \times E_{KB}) / m$

$v^2 = (2 \times 19{,}460) / 65$

$v^2 = 598.769230$

$v = 24.46976156\ ms^{-1}$

$v =$ **25 ms^{-1}** to two significant figures